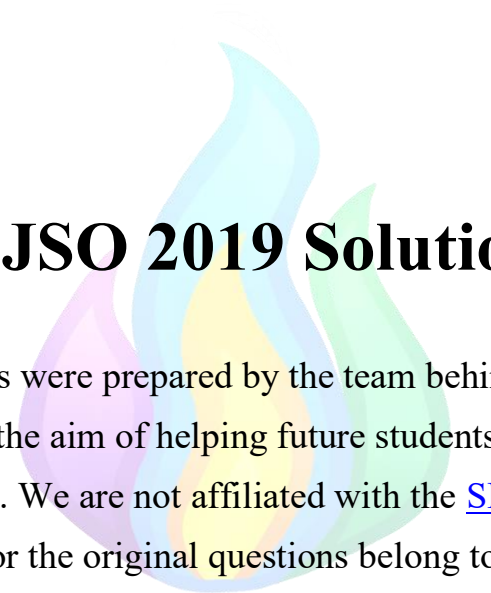




SLJSO 2019 Solutions

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<u>1</u>	Cell Biology	<u>Biomolecules</u>
<u>2</u>	Evolutionary Development	<u>Animal Classification</u>
<u>3</u>	Botany	<u>Photosynthesis</u>
<u>4</u>	Evolutionary Development	<u>Animal Classification</u>
<u>5</u>	Genetics	<u>Punnett Squares</u>
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<u>15</u>	Cell Biology	<u>Prokaryotic vs Eukaryotic Cells</u>
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<u>17</u>	Stoichiometry	<u>Mass Composition</u>
<u>18</u>	Stoichiometry	<u>Mole</u>
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<u>21</u>	Thermodynamics	<u>Ideal Gas Law</u>
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<u>23</u>	Chemical Structures	<u>Electron Configurations</u>
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Question 1

The correct answer here is option B. All biological molecules contain at least carbon, hydrogen, and oxygen. Option A is incorrect because the range of sizes is much wider when looking at all biological molecules. In biology, such harsh thresholds are typically inaccurate. Option C is inaccurate because the most common biological molecules are carbohydrates, especially cellulose. Option D is incorrect; that ratio is only true for carbohydrates, not all biological molecules. The correct answer is option B.

Correct Answer: B

Question 2

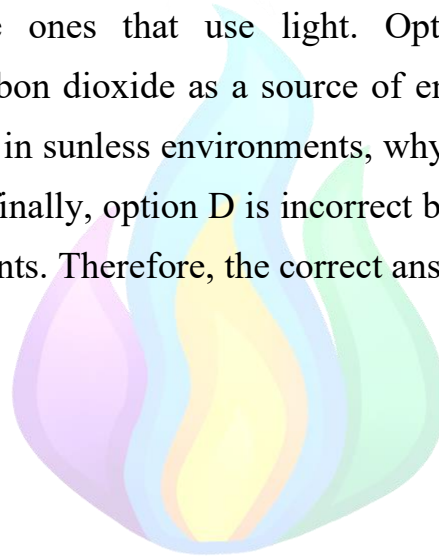
This question can be solved with general knowledge of animal kingdoms and their respiratory systems as well as the process of elimination. Earthworms, flatworms, and tapeworms use their bodies' surfaces for diffusion and respiration. We have now eliminated option D. Animals with external gills are typically aquatic amphibians; in this case that would be a tadpole. We have now eliminated option B. Crabs have internal gills, bees have tracheal systems, and dolphins have lungs. The correct answer is A.

Correct Answer: A

Question 3

To answer this question correctly, we must break the word "chemoautotroph" into "chemo" and "autotroph." Autotrophs can synthesize their own food. The chemo portion means that this organism derives its energy by fixing carbon dioxide. It is important not to confuse phytoautotrophs (i.e., plants) with chemoautotrophs (i.e., deep-sea bacteria). Although both make their own food, phytoautotrophs use sunlight, while chemoautotrophs use energy from chemical environments in sunless environments with light, like hydrothermal vents. Option A is incorrect because photoautotrophs are the ones that use light. Option B is correct because chemoautotrophs use carbon dioxide as a source of energy. Option C is incorrect because if they are found in sunless environments, why would they need a pigment to absorb sunlight? And finally, option D is incorrect because chemoautotrophs are typically bacteria, not plants. Therefore, the correct answer is B.

Correct Answer: B



Question 4

Option A is correct; earthworms use their skin for cutaneous respiration. Option B is correct because insects have an exoskeleton to prevent water loss (desiccation). Option C is false; human skin synthesizes vitamin D, not vitamin A. Option D is correct; tapeworms do not have a digestive system, so they can absorb nutrients through their skin (tegument) directly from the intestines of their host. The correct answer is C.

Correct Answer: C

Question 5

In this case, we must construct a Punnett square to identify the possible outcomes. To do that, let's assign genotypes to each parent. If black eye color is dominant to grey eye color, and a couple BOTH with black eyes can have a child with grey eyes, their genotypes must be heterozygous. After completing the Punnett square, one would find that the probability of having a child with grey eyes is 1/4. Option B is correct. The following table is helpful for checking your Punnett square:

	B	b
B	BB	Bb
b	Bb	bb

Correct Answer: B

Question 6

As we go through each statement, remember that preserving genetic diversity is important for species survival. Anything that may cause a decline in genetic diversity may lead to extinction. Option A is incorrect because global climatic changes are one of the leading causes of extinction. If an organism struggles to adapt or the changes are too severe, extinction is more likely to occur. Genetic erosion is just another way of saying genetic diversity declines. Option B is incorrect because, as mentioned previously, a decline in genetic diversity means that organisms are less likely to adapt and may lead to extinction. Option C is incorrect because accumulation of environmental pollutants makes places uninhabitable, which affects survivability. Option D is correct; it would lead to an increase in biodiversity because evolution creates more genetic variation. The correct answer is D.

Correct Answer: D

Question 7

Option A is factual because nitrogen dioxide indeed causes acid rain. Option B is false because nitrogen dioxide causes photochemical smog, not carbon dioxide. Option C is factual; CFCs, or chlorofluorocarbons, are known to cause depletion of the ozone layer. Option D is factual, because methane indeed causes global warming (in addition to carbon dioxide). Therefore, the correct answer is B since the question is looking for the incorrect answer.

Correct Answer: B

Question 8

To solve this problem, we must review Chargaff's rule: The percentage of adenine is equal to thymine, and the percentage of guanine is equal to cytosine. If we know that there is 30% guanine, and cytosine and guanine are equal, there must also be 30% cytosine. That leaves us with 40%. If adenine and thymine are equal and they add up to 40%, thymine must have a percentage of 20%. The correct answer is B.

Correct Answer: D

Question 9

One can simply use a method of elimination to successfully answer this question. Option A is incorrect because peptide bonds are found between amino acids when they form the primary structure of a protein. Option B is incorrect because disulfide bonds are used to hold proteins together to form a 3D polypeptide chain. Option C is correct; hydrogen bonds keep the alpha-helix structure of a polypeptide intact. Option D is incorrect because they help stabilize tertiary and quaternary structures by linking charged acidic and basic amino acids. Option C is correct.

Correct Answer: C

Question 10

Option A is correctly matched; water's polarity makes it behave as a very good solvent. Option B is also correctly matched because adhesion is what sticks water molecules to the inside of plant stems, while cohesion allows water molecules to stick to each other. Option C is NOT correctly matched. The high specific heat capacity of water is the reason why body temperature is very stable and does not have a wide variation. Option D is correctly matched because water expands as it freezes, which affects reservoirs in winter. Therefore, option C is correct.

Correct Answer: C

Question 11

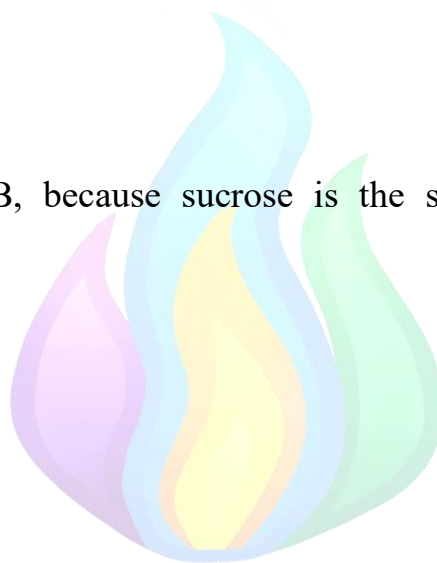
The increase in mass of precipitate directly relates to the amount of non-reducing sugars in the solution. Therefore, option A is correct because the difference between the precipitate before and after is the largest.

Correct Answer: A

Question 12

The correct answer is B, because sucrose is the strongest reducing sugar in comparison to the others.

Correct Answer: B



Question 13

The correct answer here is D. The key differentiators are that peroxisomes are used for detoxification while ribosomes are used for protein synthesis. The other options are in the wrong order.

Correct Answer: D

Question 14

- A) Yes, ATP contains adenine.
- B) Incorrect, losing one phosphate group generates 30.6 kJ, not all three
- C) Correct, ATP contains a ribose sugar
- D) Correct, ATP produces energy for endergonic reactions in cells

Correct Answer: B

Question 15

- A) Accurate, both prokaryotes and eukaryotes have cytoskeletons in the cytoplasm.
- B) Incorrect, only prokaryotes (bacteria) have the ability to fix nitrogen.
- C) Incorrect, prokaryotes do not divide via mitosis, they divide using binary fission.
- D) Incorrect, BOTH prokaryotes and eukaryotes utilize DNA for holding genetic information.

Correct Answer: A

Question 16

$$V = 2 \times 10^2 \text{ km}^3 = 2 \times 10^2 (10^4 \text{ dm})^3 = 2.0 \times 10^{14} \text{ dm}^3 = 2 \times 10^{14} \text{ L}$$

Correct Answer: A

Question 17

For each sample, the molar Fe:O ratio is $\text{Fe:O} = \frac{m_{\text{Fe}}}{55.85} : \frac{m_{\text{O}}}{16}$.

For sample A, Fe:O = 0.776, while for sample B, Fe:O = 0.711.

Since the ratios do not correspond to any common iron oxide (FeO , Fe_2O_3 , Fe_3O_4), we can conclude that the samples are mixtures of oxides, in different ratios.

Note: for sample A, $1.094 + 0.404 \neq 1.518$, but that probably is a typo.

Correct answer: C

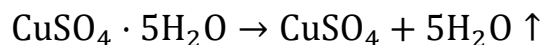
Question 18

$$n_{\text{CaCO}_3} = \frac{m_{\text{CaCO}_3}}{(40 + 12 + 3 \times 16) \frac{\text{g}}{\text{mol}}} = 0.5 \text{ mol}, \quad N = n \times 8 \text{ mol}^{-1} \times 3 = 12$$

Counting the petals, Avogadro's number is 8 mol^{-1} . In 0.5 moles, there are 4 molecules, each with 3 atoms of O, so 12 oxygen atoms in total.

Correct Answer: C

Question 19



From here, it's just basic stoichiometry:

$$n = \frac{m}{M} = \frac{1.50\text{g}}{(63 + 32 + 4 \times 16 + 5 \times (2 + 16)) \frac{\text{g}}{\text{mol}}} = 0.006\text{mol}$$

$$m = n \times M_{\text{CuSO}_4} = 0.006\text{mol} \times (63 + 32 + 4 \times 16) \frac{\text{g}}{\text{mol}} = 0.96\text{g}$$

Correct Answer: D

Question 20

Most simply, without much calculation: we need half the concentration, therefore half the volume to be the original concentration—100 ml.

A more rigorous proof would be $V = \frac{n}{0.10M} = \frac{200\text{mL} \cdot 0.05M}{0.10M} = 100\text{mL}$

Correct Answer: A

Question 21

We will use the ideal gas law:

The partial pressure of oxygen is: $P_{O_2} = P - P_{H_2O} = 722 \text{ mmHg} = 0.95 \text{ atm}$
(Dalton's law tells us that the sum of all partial pressures is the total pressure)

$$n = \frac{PV}{RT} = \frac{0.95\text{atm} \times 0.1\text{L}}{0.0821\text{Latm/mol} \times 298.15\text{K}} = 0.0039\text{mol} = 3.9 \times 10^{-3}.$$

Correct Answer: C

Question 22

From the reaction stoichiometry, $-\frac{\Delta n_{Br^-}}{5} = \frac{\Delta n_{Br_2}}{3}$

Since the rates are just the mole variations divided by time (taken as absolute values – rate of removal/formation), we have $v_{Br_2} = \frac{3}{5} v_{Br^-} = 0.30 \text{ mol s}^{-1}$

Correct Answer: C

Question 23

First of all, option C is wrong, because all the 2p orbitals are degenerate (have the same energy) – we can't depict two of the orbitals as having a higher energy.

One rule we have to follow is Hund's rule – you don't start pairing up electrons (putting two electrons in an orbital), unless all orbitals of a respective type (2p in this case) already have an electron. This means you can't have an empty 2p orbital and a dielectronic 2p orbital. Options A and C disobey this rule, so they are wrong. Option B obeys all rules, even though the last electron was placed on the middle orbital – since the 3 orbitals are degenerate, there's no difference between them, so it doesn't matter.

Correct Answer: B

Question 24

Let's look at the statements one by one:

- A. Yes, oxygen is very electronegative and H atoms bonded to it form H bonds.
- B. Even though they are much weaker than H bonds and not noticeable in water, London forces still do exist, as they would in non-polar liquids.
- C. Chemical bonds are way stronger (about 100 times) than the strongest of the intermolecular forces – H bonds
- D. It is a general rule of thumb that ionic solids dissolve in polar liquids, but there are exceptions, such as CaCO_3 , AgCl (for those interested, these exceptions arise because of the partial covalent character of the compound, or because of the exceptionally stable solid crystals)

Correct Answer: B

Question 25

Same electron structure (isoelectronic species, same number of electrons; for potassium 19-1, for argon 18, for chlorine 17+1) - look at the number of protons. More protons, stronger pull, harder to separate, so higher energy required.

Correct Answer: C

Question 26

$$P_{\text{final}}V_{\text{final}} = nRT = (n_1 + n_2)RT = n_1RT + n_2RT = P_1V_1 + P_2V_2, \text{ so, we have}$$
$$P_{\text{final}} = \frac{P_1V_1 + P_2V_2}{V_1 + V_2} = \frac{5\text{atm} \times 3\text{L} + 1\text{atm} \times 2.5\text{L}}{5.5\text{L}} = 3.18\text{atm}$$

Correct Answer: A

Question 27

As always with this kind of questions, let's analyze the statements one by one:

A. The key here is “if fully ionized”. Although acetic acid is weak and not fully ionized IF it were fully ionized, it would provide 1 mole of protons, just like 1L of 1M HCl. So, option A is correct.

B. Now we talk about the real thing – 1L of 1M HCl contains 1 mole of protons, since HCl is strong and fully ionized, while the acetic acid solution is only partially ionized, so it contains a lower amount of protons. This is also correct.

C. High pH = low proton concentration, so, using the reasoning from option B, option C is false

D. In aqueous solution, acetic acid isn't fully ionized, but, when it reacts with a strong base like NaOH, that's not relevant – the base “steals” all the protons, so it is irrelevant if the acid is strong or weak. Option D is correct.

Correct Answer: C

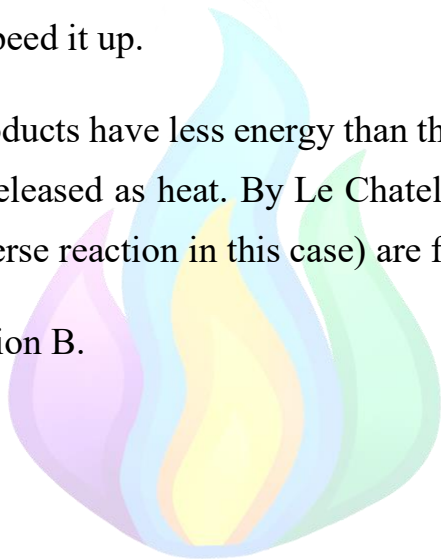
Question 28

First of all, option C is correct. Catalysts are substances that change the mechanism of a reaction in order to speed it up.

From the diagram, the products have less energy than the reactants. That energy has to go somewhere – it is released as heat. By Le Chatelier's principle, endothermic reactions (such as the reverse reaction in this case) are favored by temperature.

The false statement is option B.

Correct answer: B



Question 29

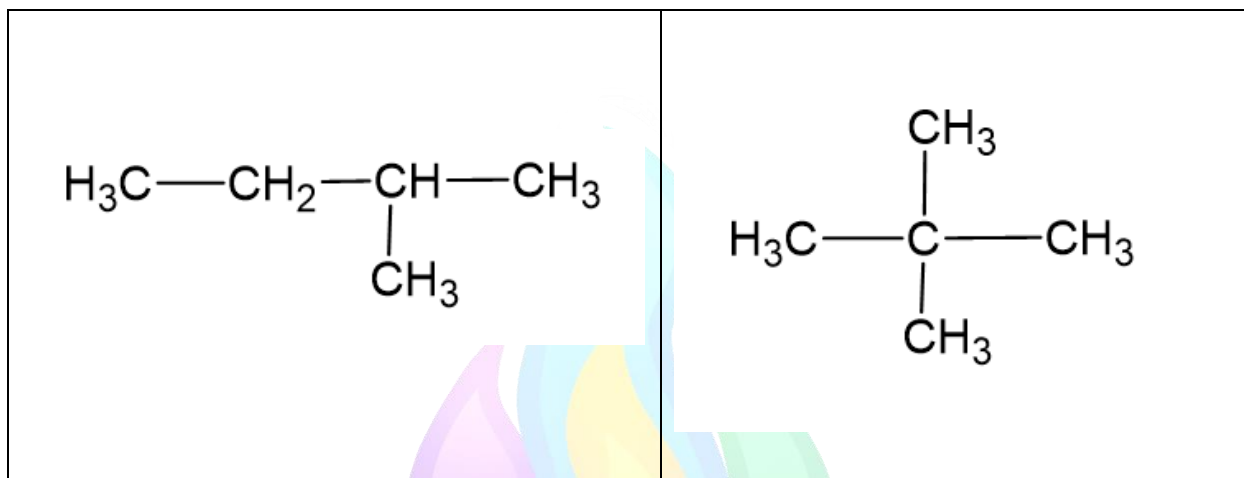
The maximum yield is 1 mol, which is $1\text{ mol} \cdot 60 \frac{\text{g}}{\text{mol}} = 60\text{g}$

The percent yield is then $\eta = \frac{55}{60} = 92\%$

Correct answer: B

Question 30

By looking at the 5-carbon chain and moving the carbons around, you can find more isomers. If you go methodically and pay attention, you won't miss any isomer. You also need to look carefully and not count an isomer twice (sometimes, if you draw an isomer the other way around, you might think it's a different compound). For n-pentane, there's two other possible isomers:



Correct Answer: C

Question 31

$$v_{\text{avg}} = \frac{s_{\text{total}}}{t_{\text{total}}} = \frac{150\text{m}}{25\text{s}} = 6 \frac{\text{m}}{\text{s}}$$

Correct Answer: A

Question 32

$$a = \frac{\Delta v}{\Delta t} = \frac{(15 - 5) \text{ ms}^{-1}}{1 \text{ s}} = 1 \frac{\text{m}}{\text{s}^2},$$

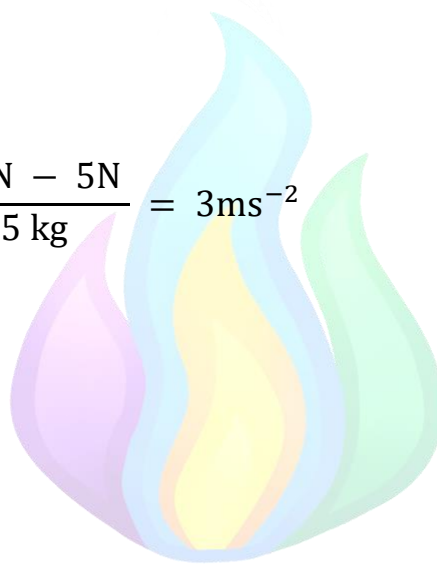
$$s = v_{\text{avg}} t = \frac{v_0 + v_f}{2} t = \frac{(15 + 5) \text{ ms}^{-1}}{2} * 10 \text{ s} = 100 \text{ m}$$

Correct Answer: D

Question 33

$$a = \frac{F}{m} = \frac{F - F_{\text{fric}}}{m} = \frac{20 \text{ N} - 5 \text{ N}}{5 \text{ kg}} = 3 \text{ ms}^{-2}$$

Correct Answer: C



Question 34

$$m = \rho V \Rightarrow V = \frac{m}{\rho}$$

$$m_{\text{oil}} = \frac{m_{\text{stone}}}{\rho_{\text{stone}}} \times \rho_{\text{oil}} = \frac{390 \text{ g}}{2.7 \text{ g cm}^{-3}} * 0.9 \text{ g cm}^{-3} = 130 \text{ g}$$

Correct Answer: A

Question 35

$$P_1 V_1 = P_2 V_2 \Rightarrow V_1 = \frac{P_2}{P_1} \times V_2 = \frac{1 \text{ atm}}{(1 + 2/10) \text{ atm}} \times 6 \text{ cm}^3 = 5 \text{ cm}^3$$

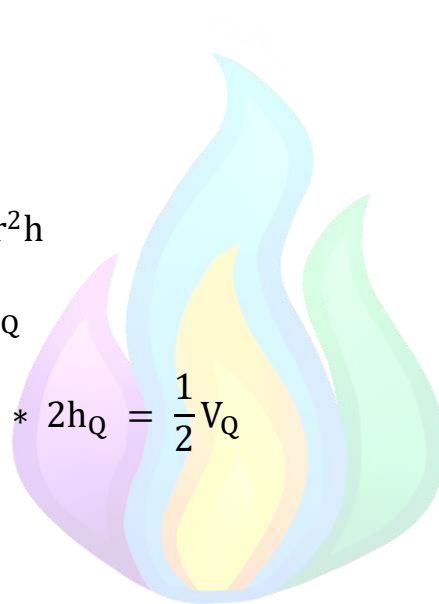
Correct Answer: C

Question 36

Volume of a cylinder = $\pi r^2 h$

$$h_P = 2h_Q \text{ and } r_P = \frac{1}{2}r_Q$$

$$V_P = \pi r^2 h = \pi \left(\frac{1}{2}r_Q\right)^2 * 2h_Q = \frac{1}{2}V_Q$$



Correct Answer: D

Question 37

If something has no acceleration, the resultant force is 0.

Correct Answer: A

Question 38

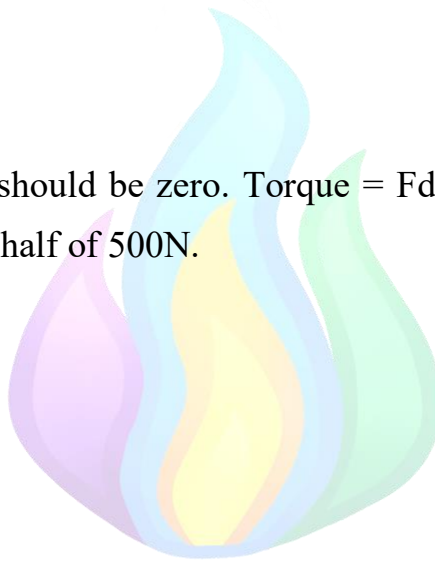
$$P = \frac{W}{t} = \frac{\text{gravitational potential energy change}}{t} = \frac{mgh}{t} = \frac{600\text{N} * 4\text{m}}{3\text{s}} = 800\text{W}$$

Correct Answer: B

Question 39

Torque around the pivot should be zero. Torque = Fd . Since 4m is twice 2m, the unknown force should be half of 500N.

Correct Answer: A



Question 40

$$Q = mc\Delta t \rightarrow \Delta t = \frac{Q}{mc} = \frac{12000\text{J}}{2\text{kg} * 400 \text{ J/kg} \cdot ^\circ\text{C}} = 15^\circ\text{C}$$

Correct Answer: A

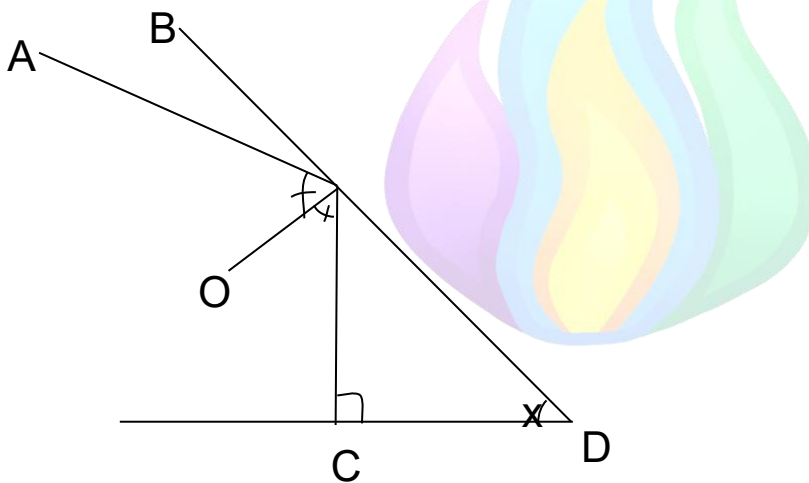
Question 41

$V = IR$. Voltage drop through B is the same as the voltage drop through C. The same goes for D and the other resistor. So, A takes the whole current, D takes half, and between B and C, more than half goes to B because it has less resistance.

Correct Answer: C

Question 42

Simple geometry, angle of incidence equals angle of reflection



From this diagram, you can see that x equals the angle of reflection equals 52°

Correct Answer: B

Question 43

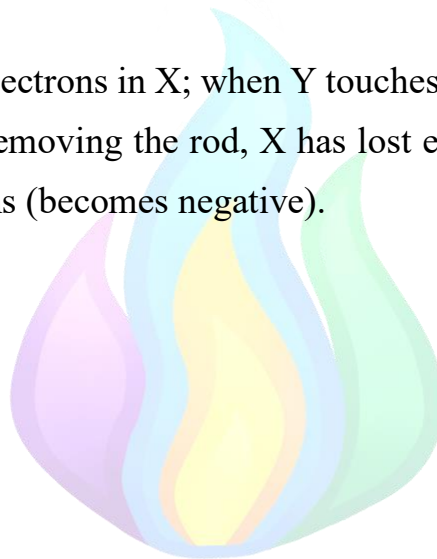
The speed of a sound in a medium is generally constant, so wavelength and frequency influence one another.

Correct Answer: A

Question 44

The negative rod repels electrons in X; when Y touches X, electrons flow from X to Y. After separation and removing the rod, X has lost electrons (becomes positive), and Y has gained electrons (becomes negative).

Correct Answer: D



Question 45

If you use the system where the index finger is the field and the middle finger is the current, then you should use the left-hand rule. However, if you use the system where the index finger is the current and the middle finger is the field, then use the right-hand rule.

NOTE:- DIFFERENT COUNTRIES USE DIFFERENT SYSTEMS. ALWAYS BE CAREFUL TO USE ONE SYSTEM THROUGHOUT A PROBLEM!

Correct Answer: B

Question 46

Snell's law: $n_1 \sin \alpha = n_2 \sin \beta$, also $\alpha = 60^\circ, \beta = 90^\circ, n_2 = 1$

Therefore $n_1 = \frac{2}{\sqrt{3}}$ (THERE IS AN ERROR IN THE TEXT, $\sin 60 = \frac{\sqrt{3}}{2}$, NOT $\sqrt{\frac{3}{2}}$)

Correct Answer: B

Question 47

In a problem like this, we should write all the terms as powers of 3.

$$\text{Right - hand side} = 3^{x-1} * 9 = 3^{x-1} * 3^2 = 3^{x+1}$$

$$\text{Left - hand side} = 3^{2x+2}$$

Since the right - hand side and the left - hand side are equal, $3^{x+1} = 3^{2x+2}$

$$x + 1 = 2x + 2$$

$$x = -1$$

Correct Answer: B

Question 48

We have to rearrange the equation given.

$$\frac{Q}{t} = \frac{P\pi a^4}{8nl}$$

$$\frac{8nlQ}{P\pi t} = a^4$$

Taking the powers of 1/4 from both sides, we get

$$\left(\frac{8nlQ}{P\pi t}\right)^{\frac{1}{4}} = a$$

Correct Answer: B



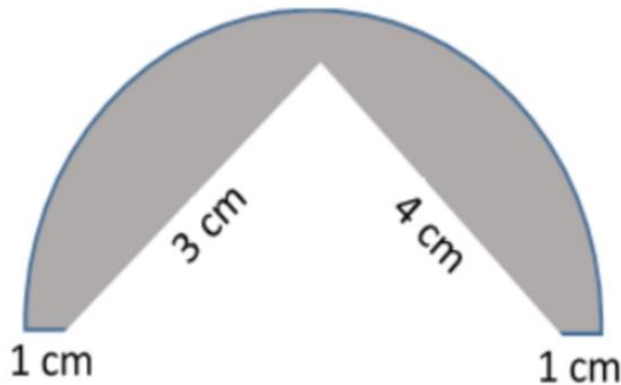
Question 49

In a question like this, you can save a lot of time if you think logically. It is quite obvious that if two lines of the form $y = mx + c$ have the same gradient but a different y-intercept, then they won't meet, but if two lines have different gradients, then they will meet.

The only option where the gradients are similar but the y-intercepts are different is option C.

Correct Answer: C

Question 50



In a question like this, it is easier to take the area of the shaded + non-shaded region and then subtract the non-shaded region.

$$\text{Area of the shaded + non - shaded region} = \frac{1}{2} \pi r^2$$

To find r , we need to find the radius of the circle. In order to do that, we need to find the length of the line joining the two 1cm lines. We can simply apply the Pythagorean theorem and find that this distance is 5 cm. Then, the diameter of the semicircle is $5\text{cm} + 1\text{cm} + 1\text{cm} = 7\text{cm}$. So, the radius is 3.5 cm.

$$\text{Area of the shaded + non - shaded region} = \frac{1}{2} \pi (3.5\text{cm})^2 = 19.24 \text{ cm}^2$$

$$\text{Area of the non-shaded region} = \frac{1}{2} \times \text{side length} = \frac{1}{2} \times 3\text{cm} \times 4\text{cm} = 6\text{cm}^2$$

$$\text{Area of the shaded region} = 19.24\text{cm}^2 - 6\text{cm}^2 = 13.24\text{cm}^2$$

Correct Answer: B